Supplementary Material 1: Notation

Notation	Description
n, p, k	number of observations, variables, embedding dimension, respectively
$\boldsymbol{X}$	$n \times p$-dimensional data
$\boldsymbol{x}_{i*}$	row i of $\boldsymbol{X}$
$\boldsymbol{x}_{*j}$	column j of $\boldsymbol{X}$
$\boldsymbol{x}^*$	new p-dimensional observation ($1 \times p$)
$\widehat{\boldsymbol{X}}$	$n \times p$-dimensional fitted values
$\boldsymbol{Y}$	$n \times k$-dimensional data (layout)
$\boldsymbol{y}_{i*}$	row i of $\boldsymbol{Y}$
$\boldsymbol{y}_{*j}$	column j of $\boldsymbol{Y}$
$\widehat{\boldsymbol{Y}}$	$n \times k$-dimensional fitted values
$\boldsymbol{y}^*$	new k-dimensional observation ($1 \times k$)
e_i	Residual of observation i, computed on p-dimensional data
P	orthonormal basis, generating a *d-dimensional* linear projection of p-dimensional data
T	true model
g	functional mapping from p-D to k-D, especially as prescribed by NLDR
$\boldsymbol{\theta}$	(Hyper-) parameters for NLDR method
r	ranges of the embedding components
$\boldsymbol{c}_{h*}^{(2)}$	coordinates of bin centers in 2-D
$\boldsymbol{c}_{h*}^{(p)}$	coordinates of bin centers in p-D
(b_1, b_2)	number of bins in each direction
(a_1, a_2)	binwidths, distance between centroids in each direction
$(s_1,\ s_2)$	starting coordinates of the hexagonal grid
q	buffer to ensure hexgrid covers data, proportion of data range, 0-1
m	number of non-empty bins
b	number of hexagons in the grid
h	hexagonal id
u_i	unique id for bin i, $i = 1, \dots, h$
H_h	set of observations in bin h
l	side length
A	area
n_h	number of points in hexagon h (bin count)
w_h	standardized number of points in hexagon h (standardized bin counts)
g	mapping from p to k dimensions
g^-	pseudo-inverse of g
$\hat{g}$	estimate of g

Table 1: Summary of notation for describing new methodology.